

RESEARCH REPORT

Customer Engagement & Product Utilization Analytics

for Retention Strategy

A Comprehensive Analytical Framework for Banking Sector
Customer Churn Prevention & Relationship Depth Optimization

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SECTION 01

Background and Context

Understanding the behavioral foundation of customer retention

Banks increasingly recognize that customer behavior and engagement — not just demographics — determine long-term retention. In the modern banking landscape, traditional indicators such as credit scores or salary brackets provide an incomplete picture of customer loyalty. Customers may appear financially strong yet still churn due to disengagement, limited product adoption, or a weak relationship with the bank.

This paradigm shift demands a deeper look at behavioral signals: how often customers interact with banking channels, how many products they use, and whether their financial commitment aligns with their activity levels. Banks that ignore these signals risk losing premium customers silently — a phenomenon known as *silent churn*.

Why Customers Churn Despite Strong Financials

Three primary behavioral drivers of churn transcend financial strength:

- **Low Engagement** — Customers who rarely interact with banking channels exhibit significantly higher churn rates regardless of balance size.
 - Active engagement is a stronger retention predictor than balance alone.
- **Limited Product Adoption** — Customers holding only one product have minimal switching costs and are easier targets for competitor acquisition.
 - Multi-product relationships create natural loyalty lock-in.
- **Weak Relationship Depth** — Transactional interactions without personalized engagement erode perceived value over time.
 - Relationship depth correlates strongly with customer lifetime value.

Strategic Design Imperatives

Understanding product utilization and engagement enables banks to design targeted initiatives across three strategic pillars:

Strategic Pillar	Focus Area	Expected Outcome
Cross-Sell Strategies	Identify under-served segments with single product usage	Increase product depth & reduce switching costs
Loyalty Programs	Reward high-engagement, multi-product customers	Reinforce retention among high-value tiers
Engagement Retention	Re-activate dormant customers before churn materializes	Reduce silent churn in premium segments

Table 1.1 — Strategic Design Pillars for Behavioral Retention

SECTION 02

Problem Statement

The data-insight gap in behavioral retention

Despite the growing availability of customer engagement and product usage data, most banks have yet to leverage this information effectively for retention strategy. A critical gap exists between *data availability* and *data-driven action*.

Gap Area	Current State	Required Capability
Behavioral Churn Drivers	No quantitative insight into behavior-churn relationships	Statistical models linking engagement metrics to churn probability
Product Depth Analysis	Unclear whether product count reduces churn meaningfully	Evidence-based product bundling strategy with measurable retention benefits
Premium Customer Risk	High-balance customers assumed to be low-churn risk by default	Identification of financially strong but behaviorally disengaged customers

Consequence of Inaction

- Retention budgets are allocated to customers who are not actually at risk
- High-value disengaged customers churn without triggering early-warning systems
- Cross-sell opportunities are missed in segments with genuine growth potential

Core Research Question:

Do customer engagement levels and product utilization depth predict churn more reliably than financial indicators such as balance or salary? And if so, which specific behavioral thresholds differentiate retained from churned customers?

SECTION 03

Project Objectives

Primary research goals and measurable outcomes

The project is structured around three primary analytical objectives, each linked to a specific methodology and measurable success criterion:

#	Objective	Analytical Method	Success Metric
01	Evaluate the relationship between engagement and churn	Chi-square test and logistic regression using Is Active Member, Num Of Products, and Exited as primary variables	Statistically significant churn rate differential between active and inactive members
02	Measure retention impact of product count and mix	Cross-tabulation of NumOfProducts vs Exited across all product tiers	Clear retention trend across product tiers (1 through 4 products)
03	Identify disengaged yet high-value customers	Segment clustering: balance x activity matrix	Quantified at-risk premium segment size with intervention priority ranking

Table 3.1 — Primary Objectives with Analytical Methods

SECTION 04

Secondary Objectives

Supporting goals for strategic impact

Beyond the core analytical objectives, this project addresses three strategic secondary goals designed to translate analytical findings into actionable business outcomes:

Support Engagement-Driven Retention Strategies

Provide data-backed evidence that enables retention teams to shift from demographic targeting to behavioral segmentation. This means defining specific engagement thresholds that trigger personalized outreach campaigns for at-risk customers.

Improve Product Bundling Decisions

Deliver insights on which product combinations correlate with the highest retention rates, enabling product managers to design smarter bundling offers and incentive structures that deepen relationship commitment.

Reduce Silent Churn Among Premium Customers

Create an early-warning detection framework for high-balance, low-engagement customers — the segment most likely to leave without any prior complaint or signal through standard satisfaction surveys.

SECTION 05

Dataset Description

Structure, variables, and analytical relevance

The analysis is based on a structured customer dataset containing 10,000 bank customer records across three geographies. Each record captures demographic, financial, and behavioral attributes, with churn status as the binary target variable.

Column	Data Type	Description	Analytical Role
CustomerId	Integer	Unique customer identifier	Record key
Surname	String	Customer surname	Non-analytical
CreditScore	Integer	Customer creditworthiness score	Control variable
Geography	Categorical	France, Spain, or Germany	Segment dimension
Gender	Binary	Male / Female	Demographic variable
Age	Integer	Customer age in years	Risk factor
Tenure	Integer	Years as a customer	Loyalty indicator
Balance	Float	Account balance amount	Financial commitment
NumOfProducts	Integer	Number of bank products held	Product depth — KEY
HasCrCard	Binary	Credit card ownership (0/1)	Engagement proxy
IsActiveMember	Binary	Activity status (0=inactive, 1=active)	Engagement — KEY
EstimatedSalary	Float	Estimated annual income	Financial capacity
Exited	Binary	Churn indicator — TARGET (0=stayed, 1=left)	Dependent variable

Table 5.1 — Dataset Column Reference (Gold rows = primary analytical variables)

Dataset Quality Considerations

- Binary variables (HasCrCard, IsActiveMember, Exited) must be validated as strictly 0/1
- NumOfProducts range should be confirmed (expected: 1–4)
- Balance field may contain zero-balance accounts requiring segment-specific treatment
- Geography distribution should be verified to avoid sampling bias in regional analysis
- No PII fields (Surname, CustomerId) are used in model training or scoring

SECTION 06

Analytical Methodology

Step-by-step framework from ingestion to insight

Phase 1	Data Ingestion & Validation
Load dataset and perform schema validation	Ensure all 13 columns are present with correct data types and expected ranges
Validate engagement and product fields	Confirm IsActiveMember, HasCrCard, NumOfProducts contain expected values
Ensure binary variable consistency	IsActiveMember and Exited must be exclusively 0 or 1 — no nulls or decimals
Confirm churn labeling accuracy	Cross-check Exited distribution against expected ~20% churn rate benchmark

Phase 2	Engagement Classification
Active Engaged Customers	IsActiveMember=1 AND NumOfProducts >= 2 — core retention group
Inactive Disengaged Customers	IsActiveMember=0 AND NumOfProducts=1 — highest churn risk segment
Active but Low-Product Customers	IsActiveMember=1 AND NumOfProducts=1 — primary cross-sell opportunity
Inactive High-Balance Customers	IsActiveMember=0 AND Balance > 75th percentile — silent churn risk group

Phase 3	Product Utilization Analysis
Churn rate by number of products	Calculate and visualize Exited rate for NumOfProducts = 1, 2, 3, 4
Single-product vs multi-product retention	Compare churn probability between NumOfProducts=1 vs NumOfProducts>=2
Product depth vs churn relationship	Determine if non-linear relationship exists (3-4 products may increase churn risk)

Phase 4	Financial Commitment vs Engagement Analysis
Balance vs activity cross-analysis	Segment matrix: 3 balance tiers x Active/Inactive to identify dominant churn quadrant

Salary-balance mismatch detection	Flag customers where EstimatedSalary >> Balance (potential competitor primary banking)
Identification of at-risk premium customers	High-balance + inactive + single-product = priority retention intervention target

Phase 5	Retention Strength Assessment
Define sticky customer profiles	Multi-product, active, long-tenure customers with credit cards — lowest churn probability
Measure churn stability across engagement tiers	Variance analysis of churn rates within each engagement classification group
Identify engagement thresholds linked to retention	Determine minimum activity and product combination for sub-10% churn probability

SECTION 07

Key Performance Indicators

Measuring engagement-driven retention outcomes

Five core KPIs have been defined to quantify retention outcomes from a behavioral and product-utilization perspective. Each KPI is directly calculable from the dataset and immediately actionable for retention strategy teams.

KPI 1	Engagement Retention Ratio Churn rate of inactive members divided by churn rate of active members. Values greater than 2.0 indicate engagement is a strong differentiator for retention. <i>Fields: IsActiveMember, Exited</i>
KPI 2	Product Depth Index Average NumOfProducts for retained customers minus average for churned customers. Positive values confirm that deeper product relationships reduce churn probability. <i>Fields: NumOfProducts, Exited</i>
KPI 3	High-Balance Disengagement Rate Churn rate among customers in the top 25% of Balance who are also inactive. This directly quantifies silent churn risk in the premium customer segment. <i>Fields: Balance, IsActiveMember, Exited</i>
KPI 4	Credit Card Stickiness Score Churn rate difference between HasCrCard=0 and HasCrCard=1 customer groups. A positive score validates credit card ownership as a key retention anchor product. <i>Fields: HasCrCard, Exited</i>
KPI 5	Relationship Strength Index Composite score: $(Is\ Active\ Member \times 0.4) + (NumOfProducts/4 \times 0.35) + (Tenure/max_tenure \times 0.25)$. Higher scores predict lower churn probability. <i>Fields: IsActiveMember, NumOfProducts, Tenure, Exited</i>

SECTION 08

Streamlit Web Application Requirements

Interactive dashboard specifications and user capabilities

The project deliverable includes a Streamlit web application providing interactive analytics for both technical analysts and business stakeholders. The dashboard is organized into four core modules with configurable filters and real-time visualization updates.

Core Analytical Modules

Module 1	Engagement vs Churn Overview
✓	Active vs inactive churn rate comparison with statistical significance testing
✓	Geography-stratified engagement analysis across France, Spain, Germany
✓	Time-series view of engagement trends segmented by customer tenure cohort
✓	Engagement Retention Ratio KPI card with industry benchmark comparison
Module 2	Product Utilization Impact Analysis
✓	Churn rate bar chart segmented by NumOfProducts (1 through 4)
✓	Single-product vs multi-product side-by-side retention comparison
✓	Product Depth Index trend with confidence intervals
✓	Cross-sell opportunity matrix: active customers with single-product usage
Module 3	High-Value Disengaged Customer Detector
✓	Scatter plot: Balance vs activity status with churn outcome overlay
✓	Priority intervention customer table with CSV export functionality
✓	High-Balance Disengagement Rate KPI with configurable threshold alerts
✓	Salary-balance mismatch flag indicator per customer segment
Module 4	Retention Strength Scoring Panel
✓	Relationship Strength Index distribution histogram by customer tier
✓	Top and bottom decile customer profile comparison dashboard
✓	Churn probability heat map across engagement x product depth grid



Sticky customer archetype definition and segment size quantification

User Control Capabilities

Control Type	Functionality
Engagement Filters	Toggle between active/inactive/all member status with real-time chart updates
Product Count Sliders	Select minimum and maximum NumOfProducts range (1–4) for filtered analysis
Balance & Salary Thresholds	Configurable percentile sliders for premium customer segment definition

SECTION 09

Deliverables and Submission

Project outputs and stakeholder artifacts

This project produces three distinct deliverables, each designed for a specific audience and purpose within the retention strategy lifecycle.

D1 Research Paper

A comprehensive analytical document covering exploratory data analysis, statistical findings, behavioral segmentation insights, and evidence-based retention recommendations.

- Full EDA with visualizations and statistical summaries
- Hypothesis testing results for engagement-churn relationship
- Product utilization findings with retention implications
- Behavioral segment profiles and cross-sell recommendations

D2 Streamlit Dashboard

A live interactive analytics application enabling real-time exploration of engagement metrics, product utilization patterns, and customer risk scores through configurable filters.

- Four analytical modules with interactive controls
- KPI scorecard with trend indicators and benchmark comparisons
- At-risk customer export with intervention priority ranking
- Mobile-responsive layout for stakeholder presentations

D3 Executive Summary

A concise, visually-rich briefing document for government and senior banking stakeholders, translating analytical findings into strategic policy recommendations without requiring technical expertise.

- 3–5 page non-technical summary of key findings
- Regulatory-relevant insights on customer behavior patterns
- Prioritized action recommendations with expected impact estimates
- Industry benchmark comparisons where available

Conclusion

Reframing churn through behavior and relationship strength

This project represents a fundamental reorientation of customer churn analysis — from a demographic and financial lens to a behavioral and relationship-strength framework. By centering the analysis on engagement activity and product utilization depth, it addresses the most actionable levers banks have for influencing retention outcomes before customers reach the point of no return.

Traditional churn models that rely heavily on credit scores, salary brackets, or geographic proxies produce interventions that are misaligned with actual customer intent. In contrast, behavioral signals — whether a customer is active, how many products they hold, and whether their financial engagement matches their profile — are both more predictive and more actionable.

Key Contributions of This Framework

Contribution	Strategic Value
Behavioral Segmentation	Classifies customers by engagement activity and product-holding profiles rather than static demographics, enabling more precise targeting of retention interventions
Silent Churn Detection	Creates an early-warning system that flags high-value customers at hidden churn risk — those who disengage financially before any complaint or survey signal is triggered
Product Strategy Alignment	Connects product bundling and cross-sell decisions directly to measurable retention outcomes, giving product managers evidence-based criteria for offer design
Actionable KPI Framework	Delivers five calculable metrics — Engagement Retention Ratio, Product Depth Index, High-Balance Disengagement Rate, Credit Card Stickiness Score, and Relationship Strength Index — that retention teams can track operationally
Stakeholder Communication	Translates complex behavioral analytics into executive-ready strategic guidance, bridging the gap between technical findings and boardroom-level decision-making.

Ultimately, the value of this project lies not only in what it measures but in what it enables: a shift from reactive churn response to proactive engagement management. When banks understand which customers are disengaging before they leave, and why, they gain the strategic advantage needed to retain relationships that would otherwise be silently lost.

Project Vision

To equip banking retention teams with a behavioral intelligence framework that identifies at-risk customers earlier, targets interventions more precisely, and builds product relationships that make customers genuinely harder to lose.